

Private SLM & Sensitive-Data AI Roadmap

PRIVATE SLM & SENSITIVE-DATA AI ROADMAP

Goal

Build the skills to design, fine-tune, deploy, and secure small language models (SLMs) for narrow business workflows where sensitive information such as PII/PHI should remain inside a controlled environment.

Core architecture

Private data -> Retrieval/RAG -> Local SLM -> AI workflow -> Controlled user/application

Important principle

Fine-tuning teaches a model how to perform a task; RAG supplies changing or private information at inference time. For sensitive business data, avoid assuming that fine-tuning on raw PII/PHI is necessary. A private SLM + RAG architecture can keep sensitive records outside model weights.

PHASE 1 — AI / ML FOUNDATIONS

Learn:

- Python for ML
- PyTorch fundamentals
- Neural networks
- Transformers
- Tokens and tokenization
- Embeddings
- Attention
- Context windows
- Pretraining vs. instruction tuning
- Inference and model serving

Target outcome:

Understand what happens between an input and an SLM's output.

PHASE 2 — SMALL LANGUAGE MODELS

Learn:

- What makes an SLM different from a larger LLM
- Open-weight models
- Instruction tuning
- Supervised fine-tuning (SFT)
- Dataset preparation
- Chat templates

- Model evaluation

Primary resource:

Hugging Face Smol Course

<https://huggingface.co/learn/smol-course/unit0/1>

Additional:

<https://github.com/huggingface/smol-course>

Target outcome:

Fine-tune an existing small model for a narrow task.

PHASE 3 — PARAMETER-EFFICIENT FINE-TUNING

Learn:

- PEFT

- LoRA
- Adapters
- Quantization-aware workflows
- Training/evaluation splits
- Hyperparameters
- Checkpoints
- Model merging

Primary resource:

Hugging Face PEFT

<https://huggingface.co/docs/peft/index>

Target outcome:

Adapt a pretrained SLM without retraining every parameter.

PHASE 4 — RAG & PRIVATE KNOWLEDGE

Learn:

- Document ingestion
- Chunking
- Embeddings
- Vector databases
- Similarity search
- Metadata filtering
- Retrieval pipelines
- Context construction

- RAG evaluation
- When to use RAG vs. fine-tuning

Target outcome:

Build an application where the model can answer from private documents without putting those documents into model weights.

PHASE 5 — LOCAL / PRIVATE INFERENCE

Learn:

- Local model execution
- Quantization
- CPU/GPU inference
- Model formats
- Model servers
- REST APIs
- Containers
- Private networking
- Resource/cost optimization

Resource:

llama.cpp

<https://github.com/ggml-org/llama.cpp>

Target outcome:

Run an SLM locally and expose it to an application through a private interface.

PHASE 6 — AI WORKFLOW ENGINEERING

Learn:

- Structured outputs
- Tool calling
- Workflow orchestration
- Validation
- Human-in-the-loop approvals
- Retries/fallbacks
- State management
- Observability
- Audit trails

Target outcome:

Turn the model into a reliable business process rather than a chatbot.

PHASE 7 — SECURITY & GOVERNANCE

Learn:

- Authentication
- Authorization/RBAC
- Encryption at rest/in transit
- Secrets management
- Network isolation
- Data minimization
- PII/PHI detection
- Audit logging
- Retention/deletion policies
- Prompt-injection defenses
- Output validation
- Model evaluation
- Human approval
- Incident response
- AI governance

Primary resource:

NIST AI Risk Management Framework

<https://www.nist.gov/itl/ai-risk-management-framework>

AI Risk Management Framework Resource Center:

<https://airc.nist.gov/>

Target outcome:

Understand how to design the entire AI system—not just the model—with security and governance in mind.

PHASE 8 — CAPSTONE

Build a complete private AI application using:

- Open-weight SLM
- LoRA/PEFT fine-tuning
- Private RAG
- Local model serving
- API/application layer
- Authentication and authorization
- Audit logging
- Evaluation suite
- Containerized deployment

- Monitoring

Suggested progression

1. Run an SLM locally.
2. Build a basic inference API.
3. Build a RAG application using synthetic/non-sensitive data.
4. Fine-tune the SLM on a narrow task using LoRA.
5. Compare base vs. fine-tuned model.
6. Add structured outputs and validation.
7. Add authentication and authorization.
8. Add audit logging.
9. Containerize the system.
10. Deploy it in a private environment.
11. Evaluate accuracy, hallucination rate, latency, cost, and security controls.

IMPORTANT BUSINESS IDEA TO REMEMBER

“We build private AI systems that allow organizations to automate sensitive workflows without sending their data to external AI providers.”

Potential positioning:

Do not sell only “an SLM.” Sell a complete private AI workflow.

Potential target industries:

- Healthcare
- Insurance
- Finance
- Legal
- Government
- Accounting
- Human resources
- Manufacturing

Potential architecture:

Customer environment

- > Sensitive database/documents
- > Access-controlled retrieval/RAG
- > Task-specific local SLM
- > AI workflow engine
- > Human/application output

Commercial insight:

The SLM is a component. The value proposition is secure automation of a useful business workflow while maintaining control over sensitive data.

IMPORTANT TECHNICAL CAVEAT

Running a model locally does not automatically make a system HIPAA-compliant or otherwise secure. The database, application, logs, backups, network, model server, access controls, encryption, and operational processes must also be secured and governed according to the applicable requirements.

KEY RESOURCE LINKS

1. Hugging Face Smol Course:

<https://huggingface.co/learn/smol-course/unit0/1>

2. Smol Course GitHub:

<https://github.com/huggingface/smol-course>

3. Hugging Face PEFT:

<https://huggingface.co/docs/peft/index>

4. llama.cpp:

<https://github.com/ggml-org/llama.cpp>

5. NIST AI Risk Management Framework:

<https://www.nist.gov/itl/ai-risk-management-framework>

6. NIST AI Resource Center:

<https://airc.nist.gov/>